

10(a)(i)

any **three** from:

- 1 as the, blood glucose / glucose concentration, increases the, blood insulin / insulin concentration, increases ; **A** positive correlation
- 2 data quote with unit ;

data quote for mp1

time of day ±1.5 min	blood glucose conc / mmol dm ⁻³ ±0.025	time of day ±1.5 min	blood insulin conc / pmol dm ⁻³ ±1
07.00	4.3	07.15	30
08.00	6.2	08.00	280

data quote for mp5

time of day ±1.5 min	blood glucose conc / mmol dm ⁻³ ±0.025	time of day ±1.5 min	blood insulin conc / pmol dm ⁻³ ±1
08.00	6.2	08.00	280
09.00	4.45	09.00	120

- 3 increase in, blood glucose / glucose concentration, **causes** release of insulin (from pancreas) ; **ora**
- 4 insulin stimulates the conversion of glucose to glycogen / glycogenesis
or
insulin increases permeability (of liver / muscle) cells to glucose / AW ;
- 5 insulin causes, blood glucose / glucose concentration, to return back to set point ;
- 6 *ref. to* negative feedback ;

3

Question	Answer	Marks
10(a)(ii)	any three from: max 2 if only glucose or insulin curve mentioned 1 ref. to delay before both graphs increase / AW ; 2 peaks for both would be lower ; 3 both curves would take longer to decrease ; 4 time is needed for starch to be, broken down / converted (to glucose) ; 5 to glucose ;	3
10(b)	any four from: 1 ref. to adenylyl cyclase ; 2 formation of, cyclic AMP / cAMP ; 3 cAMP acts as a second messenger ; 4 activation of (protein) kinase ; 5 enzyme cascade ; 6 amplification of signal ; 7 glycogenolysis / gluconeogenesis / described ; 8 glucose released into blood ;	4